MIL-P-13893E (MU)
1 July 1969
SUPERSEDING
MIL-P-13893D
1 July 1965

MILITARY SPECIFICATION

FIRING DEVICE, DEMOLITION, DELAY TYPE, M1 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for, loading, assembling, and packing for one type of firing device designated as Firing Device, Demolition, Delay Type, M1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-P-2497 - Primer, Percussion, M27
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

8846784 - Firing Device, Demolition, Delay Type, M1 Assembly
8861384 - Box, Set-Up, Packing, Ammunition, for Firing Device, Demolition, Delay Type, M1
8861385 - Box, Wire-Bound, Packing, Ammunition for Firing Device, Demolition, Delay Type, M1

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8846784 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.-The assemblies shall comply with all requirements specified on Drawing (dwg.) 8846784 and with all requirements specified in applicable specifications.

3.3 Leakage.-The ampoule assembly shall show no evidence of leakage or microscopic cracks when subjected to a vacuum of 25 inches of mercury, minimum, for one minute, when tested as specified in 4.4.1.

3.4 Spring embrittlement.-Springs manufactured for use in this assembly shall be free from embrittlement as evidenced by not fracturing when extended to the point of minimum permanent distortion, when tested as specified in 4.4.3.

3.5 Tension wire cleanliness.-The surface of the tension wire shall be free of all foreign matter and shall be kept in that condition through all production and assembly processes (see 4.4.5).

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, corrision products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall be parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

3.7 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.1.2. The first article sample shall consist of 80 complete assemblies and 25 complete sets of parts (i.e.; 25 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect production uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components and assemblies will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component, subassembly or assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any component or subassembly in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Percussion primers from not more than one lot interfix number from one manufacturer.
b. Delay charge solution from not more than one batch or homogeneous blend of batches.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Tube (see dwg. 8846775 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Wall thickness, minimum (min.)	Gage	01001
102.	Inside diameter, maximum (max.)	Gage	01004
103.	Parallelism of ends	Gage	01005
Minor:	AQL 1.00 percent		
201.	Total length, min	Gage	01002
202.	Evidence of poor workmanship, (see 3.6)	Visual	01003

4.3.2.2 Tube, outer (see dwg. 8846776 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 20261O-18T02:33Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Total length, max.....	Gage	02001
202.	Inspection hole missing.....	Visual	02004
203.	Evidence of poor workmanship, (see 3.6).....	Visual	02002

4.3.2.3 Pin, firing (see dwg. 8846774 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Profile of point.....	Gage	03001
Minor:	AQL 1.50 percent		
201.	Total length, min.....	Gage	03002
202.	Large outside diameter, max.....	Gage	03003
203.	Protective finish improper or incomplete.....	Visual	03004
204.	Evidence of poor workmanship, (see 3.6).....	Visual	03005

4.3.2.4 Coupling, base (see dwg. 8846780 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Flash hole obstructed.....	Visual	04001
102.	Concentricity of threads with outside diameter of primer cavity end.....	Gage	04006
103.	Pitch diameter of threads, min..	Gage	04007
104.	Major diameter of threads, min..	Gage	04008

Source: https://assist.dla.mil -- Downloaded: 20261O-18T02:33Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Total length, max.	Gage	04002
202.	Outside diameter of flange	Gage	04003
203.	Countersink missing	Visual	04004
204.	Evidence of poor workmanship, (see 3.6)	Visual	04005

4.3.2.5 Spring, plunger (see dwg. 8846781 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Load at test height	Test	05004
102.	Solid length	Gage	05005
103.	Protective finish missing or inadequate	Visual	05001
Minor:	AQL 1.00 percent		
201.	End not square	Visual	05002
202.	Evidence of poor workmanship, (see 3.6)	Visual	05003

4.3.2.6 Ampoule assembly (see dwg. 8846783 covering a detail of dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Improper color coding	Visual	06001
Major:	AQL 0.40 percent		
101.	Level of delay charge	Scale	06002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	06003

4.3.2.7 Assembly, prior to securing base coupling (see dwg. 8846784).

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:33Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Primer missing	Visual	07003
102.	Sealing compound on primer	Visual	07004
Minor:	None defined		

4.3.2.8 Assembly (see dwg. 8846784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Safety strip improperly assembled	Visual	08001
102.	Safety strip color code improper	Visual	08002
103.	Threads damaged	Visual	08003
104.	Ampoule loose	Manual	08004
105.	Base coupling loose	Manual	08005
106.	Tube loose	Manual	08006
107.	Fixing screw loose	Manual	08007
108.	Shipping cup missing or improperly assembled	Visual-Manual	08008
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship, (see 3.6)	Visual	08009

4.3.2.9 Set-up box, unsealed (see dwg. 8861384).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Temperature correction chart missing	Visual	09001
102.	Device improperly positioned	Visual	09002
Minor:	None defined.		

Source: https://assist.dla.mil -- Downloaded: 20261O-18T02:33Z
Check the source to verify that this is the current version before use.

4.3.2.10 Set-up box, sealed (see dwg. 8861384).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Corners not blunted	Visual	10001
102.	Seal insecure	Manual	10002
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	10003

4.3.2.11 Barrier bag, sealed (see dwg. 8861384).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Air not exhausted	Visual	11001
102.	Sealed not permitting subsequent seals	Visual	11002
103.	Seal inadequate or bag damaged	Visual-Manual	11003
Minor:	AQL 0.40 percent		
201.	Marking missing, misleading or unidentifiable	Visual	11004

4.3.2.12 Box, packing, sealed (see dwg. 8861385).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Board broken	Visual	12001
102.	FSC number missing or misleading	Visual	12002
103.	Contents exposed	Visual	12003
104.	Cover loose	Manual	12004

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.0 percent		
201.	Marking missing, misleading or unidentifiable.....	Visual	12005
202.	Car seal missing, unsealed or improperly positioned.....	Visual	12006

4.3.3 Testing.

4.3.3.1 Leakage (see 3.3) - Major defect - Code No. 13001.- Sampling plan for this characteristic shall be in accordance with MIL-STD-105, using code letter M and an AQL of .040 percent. Testing shall be performed as specified in 4.4.1 using test equipment specified in 4.3.4.

4.3.3.2 Crush test of tube (see dwg. 8846775 covering a detail of dwg. 8846784) - Major defect - Code No. 14001.- A sample of 25 tubes shall be selected from each inspection lot for this test. Should one or more units of the sample exhibit a crack, the lot shall be rejected not subject to retest. Testing shall be as specified in 4.4.2 using the applicable equipment of 4.3.4.

4.3.3.3 Spring embrittlement (see 3.4) - Major defect - Code No. 15001.- A sample of 10 plunger springs shall be selected from each heat treatment lot for this test. If any spring fails to meet the requirement, the heat treatment lot shall be rejected. The test shall be performed as specified in 4.4.3.

4.3.3.4 Spring load at assembled height (see dwg. 8846781 covering a detail of dwg. 8846784) - Major defect - Code No. 05006.- A sample of 25 springs shall be selected from each heat treat lot for this determination. Failure of one or more units in meeting the applicable drawing requirement shall reject the heat treat lot represented by the sample. The test shall be performed as specified in 4.4.4.1, using test equipment in accordance with 4.3.4.

4.3.3.5 Cleanliness of tension wire (see 3.5) - Major defect - Code No. 16001.-A sample of 25 firing pin and tension wire subassemblies shall be selected from each 4 hours production for this test. Testing shall be as specified in 4.4.5. Failure of one or more units of the sample in meeting the requirement shall reject the product represented subject to cleaning and resubmission to test.

4.3.3.6 Functioning (see dwg. 8846784) - Major defect - Code No. 08010.-A sample of 80 firing device assemblies shall be selected from each lot. Forty (40) assemblies shall be tested

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

with the assembly held in a horizontal position with the tension wire above the ampoule. The remaining forty (40) assemblies shall be tested with the assembly held vertical with the primer end down. The tests shall be performed as detailed in 4.4.6 using test equipment specified in 4.3.4. The lot shall be rejected if, in either the horizontal or vertical tests, the mean value of functioning time "$\bar{x}$" plus or minus 2.2(K) multiplied by the standard deviation does not fall within the limits as specified on dwg. 8846784 (see 6.5). In addition, the lot shall be rejected if one or more items of the sample for the black item functions in less than four (4) minutes; for the red item functions in less than ten (10) minutes; for the white item, functions in less than thirty-seven (37) minutes; for the green item, functions in less than seventy (70) minutes; for the yellow item, in less than one hundred and sixty (160) minutes; or for the blue item, in less than four hundred and fifty (450) minutes.

4.3.3.7 After three consecutive lots have met the criteria of 4.3.3.6, a sample of 50 firing device assemblies shall be selected from each lot. The test shall be performed to 25 samples each as specified in 4.3.3.6 and as detailed in 4.4.6 using test equipment specified in 4.3.4. The lot shall be rejected if in either the horizontal or vertical tests the mean value of functioning time "$\bar{x}$" plus or minus 2.07(K) multiplied by the standard deviation does not fall within the limits as specified on dwg. 8846784 (see 6.5). In addition, the lot shall be rejected if one or more items of the sample for the black item, functions in less than four (4) minutes; for the red item, functions in less than ten (10) minutes; for the white item, functions in less than thirty-seven (37) minutes; for the green item, functions in less than seventy (70) minutes; for the yellow item, in less than one hundred and sixty (160) minutes; or for the blue item, in less than four hundred and fifty (450) minutes.

4.3.3.8 Check test for possible deterioration (see MIL-P-2497.) - If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into the firing device exceeds two years, or if the primers have been subjected to adverse conditions for any period of time since previous tests, the primer lot shall be subjected to any must satisfactorily pass the check test for possible deterioration required by the primer specification immediately before the primer lot is assembled into the firing device. These tests shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the firing device. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or the contractor assembling

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

the primers into the firing device, and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

4.3.3.9 Firing pin, released prematurely or not retracted, (see dwg. 8846784) - Major defect - Code No. 08011.- This test shall be performed 100 percent. If any assembly shows evidence of a premature firing pin release or a firing pin not properly retracted, or a firing pin resting on the safety strip, it shall be classed defective and removed from the lot. Testing shall be performed as specified in 4.4.7.

4.3.3.10 Heat seal test of barrier bag (see dwg. 8861384) - Major defect - Code No. 17001.- Sampling, acceptance criteria and test procedures for this test shall be as required in MIL-P-116.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET 8846784 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection Equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering, and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test method and procedures.

4.4.1 Leakage.-The ampoule assembly shall be placed in the applicable test equipment, the required vacuum applied and held for the specified period. The vacuum shall then be reduced, slowly, to atmospheric pressure by admission of ammonia gas to the test chamber. Observation shall be made for blue discoloration or for a blue line, or lines, in the ampoule which indicate a defective.

4.4.2 Crush test of tube.-The tube shall be assembled into the applicable test equipment, the required load applied at a rate not exceeding 1/8 inch per minute until the tube wall has flattened against itself and observation made for a crack or cracks in each edge. Any unit exhibiting a crack shall be classed defective.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:33Z
Check the source to verify that this is the current version before use.

4.4.3 Spring embrittlement.-The spring shall be repeatedly stretched with increasing loads until it does not return to its original free height. That point shall indicate minimum distortion. Any unit which fractures prior to extension to minimum distortion shall be classed defective.

4.4.4 Spring load at assembled height.- The springs shall be maintained at the specified height for 24 hours minimum. The springs shall then be tested for compliance with the load requirement.

4.4.5 Tension wire cleanliness.- The wire of the firing pin and tension wire subassembly shall be dipped in water maintained at 68 plus or minus 5 degrees Fahrenheit for 10 seconds, maximum, and shall then be slowly removed from the water. Observation shall be made for "no break" water film coverage using adequate magnification. Any unit exhibiting a broken film shall be classed defective.

4.4.6 Functioning.- The firing device shall be assembled to the applicable test equipment and the tube crushed so as to break the ampoule. The time shall be measured from when the tube is crushed until the primer is functioned. The test shall be performed at 77 degrees Fahrenheit plus or minus 2 degrees Fahrenheit.

4.4.7 The firing device after a minimum of eight (8) days of its manufacture and final assembly and before being unit packaged (see dwg. 8861384), shall be visually inspected. Observation shall be made to see if the firing pin is resting on the safety strip, not properly retracted or prematurely released. Any unit showing any evidence thereof shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.2 Level A - The firing devices shall be preserved and packaged in accordance with dwg. 8861384.

5.2 Packing

5.2.1 Level A - The packaged devices shall be packed in accordance with dwg. 8861385.

5.3 Marking - Marking shall be in accordance with dwgs. 8861384 and 8861385.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T02:33Z
Check the source to verify that this is the current version before use.

5.4 Temperature correction table.- One copy of the temperature correction table detailed on figure 1 shall be packed in each unit package (see dwg. 8861384). The sentence preceding the table shall be in red letters. The other letters and digits shall be in black. The overall size of the table shall be not more than 4 inches wide by 5 inches long.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Check test for deterioration.- The contracting officer shall arrange to have the contractor reimbursed for any expense incurred in performing the check test for deterioration.

6.5 The standard deviation may be computed by using the following formula:

$$\text{Standard Deviation } (s) = \sqrt{\frac{\Sigma(\bar{x}-x)^2}{N-1}}$$

where:

$\Sigma(\bar{x}-x)^2$ equals the sum of squared deviations of each item from the mean.

N-1 equals one less than the sample size.

$\bar{x}$ equals mean or arithmetic average or sum of all results divided by the number of results.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-046

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:33Z
Check the source to verify that this is the current version before use.

FIGURE I

Destroy this table after it has served its purpose.

TEMPERATURE CORRECTION TABLE

t F	BLACK		RED		WHITE	
	CM	ST	CM	ST	CM	ST
-25			8.5 hr	3.3 hr	3 day	1.3 day
0	8 hr	2.5 hr	45 min	20 min	17.5 hr	8 hr
plus 25	36 min	16 min	25 min	11 min	5.5 hr	2.5 hr
50	15 min	7 min	17 min	8 min	2 hr	55 min
75	9 min	4 min	15 min	7 min	1 hr	27 min
100	5 min	2.0 min	8 min	3.5 min	32 min	14 min
125	4 min	1.5 min	5 min	2 min	20 min	9 min
150	3 min	1 min	4 min	1.5 min	15 min	6 min

t°F	GREEN		YELLOW		BLUE		t°C
	CM	ST	CM	ST	CM	ST	
-25							-32
0	2.6 day	1.2 day	8.5 day	3.8 day	23 day	10 day	-18
plus 25	17 hr	8 hr	2.0 day	20 hr	5.0 day	2.2 day	-4
50	6 hr	2.7 hr	14 hr	6.0 hr	1.3 day	14 hr	plus 10
75	2.5 hr	70 min	5.5 hr	2.5 hr	11.5 hr	5 hr	24
100	70 min	30 min	2.5 hr	65 min	5.2 hr	2.3 hr	38
125	35 min	15 min	80 min	36 min	2.5 hr	1.1 hr	52
150	20 min	9 min	46 min	21 min	80 min	36 min	66

CM - When two pencils are used in the same charge, the CM is the most likely timing. When only a single pencil is used, the value should be increased by about 15 percent.

ST - The ST is a reasonably safe time. Timings shorter than the ST should not occur more often than once in a thousand trials. Red pencils should not be used below 0 degrees F., nor Black pencils below 25 degrees F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:33Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:33Z
Check the source to verify that this is the current version before use.